Roll No. ....

TEE–301

B. TECH. (EE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

ELECTRICAL CIRCUIT ANALYSIS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain KVL and Supermesh. Find the currents I_1 and I_2 at the network shown in Fig. 1. (CO1, 2)

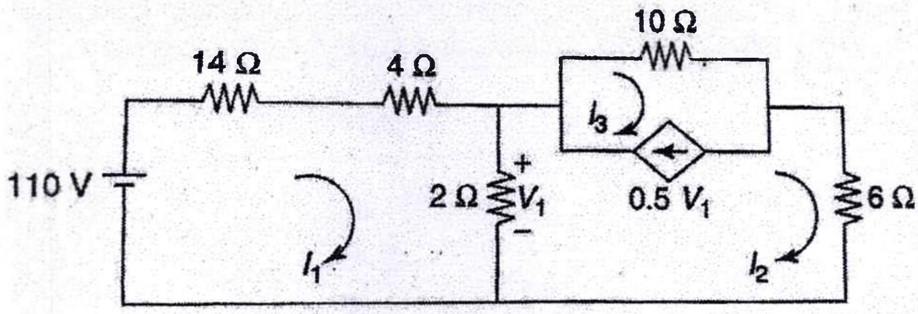

Fig. 1

P. T. O.

(b) Determine the current through the 10 ohm resistor in the Fig. 2 using superposition theorem. (CO1, 2)

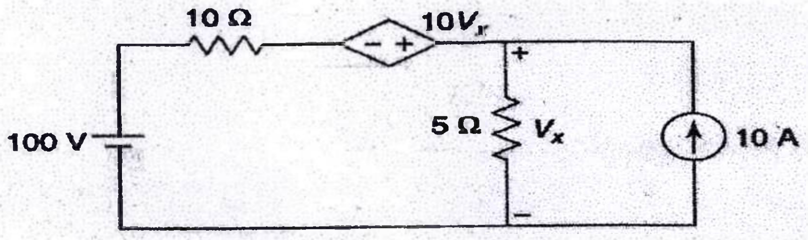

Fig. 2

(c) For the network shown in Fig. 3, find the value of R_L for maximum power transfer. Also, find the maximum power. (CO1, 2)

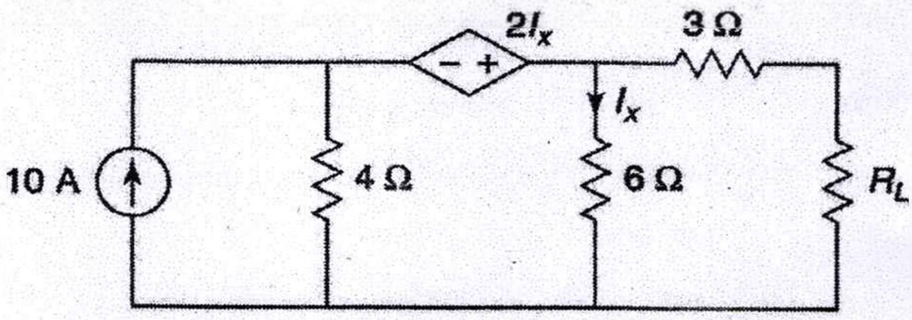

Fig. 3

2. (a) Define the following terms : Planer graph, Connected graph, Rank of a graph, Tree, Co-tree, Directed graph with the help of example. (CO2, 3)

(b) The graph of a network is shown in Fig. 4. Write the (i) incidence matrix, (ii) tieset matrix, and (iii) f – cutset matrix. (CO2, 3)

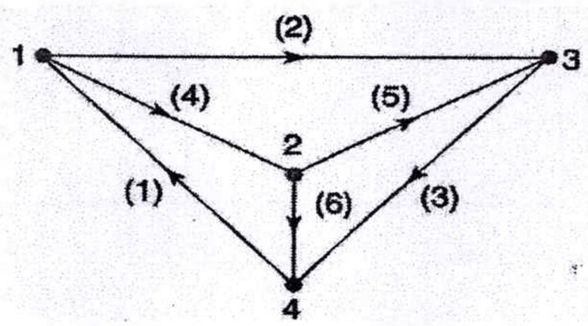

Fig. 4

(c) A balanced delta-connected load of impedance $(8 - j6)$ ohms per phase is connected to a three-phase, 230 V, 50 Hz supply. Calculate (i) power factor, (ii) line current, and (iii) reactive power. (CO2, 3)

3. (a) The network shown in Fig 5 has acquired steady-state with the switched closed at $t < 0$. At $t = 0$, the switched is opened. Obtain $i(t)$ for $t > 0$ using Laplace transform. (CO4, 5)

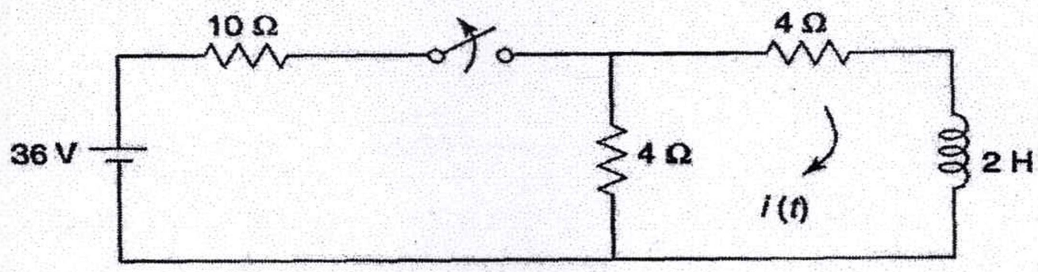

Fig. 5

(b) Find the current $i(t)$ for the circuit shown in Fig 6. if the voltage source is $v(t) = 5e^{-2t}u(t)$ and $v_c(o-) = 0$. (CO4, 5)

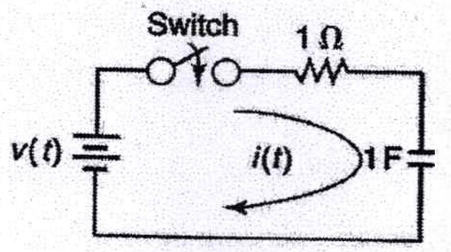

Fig. 6

(c) Find the inverse Laplace transform of the function : (CO4, 5)

$$F(s) = \frac{12}{(s+2)^2 (s+4)}$$

4. (a) Test result for a two-port network are

(i) $I_1 = 0.1\angle 0°A$, $V_1 = 5.2\angle 50°V$, $V_2 = 4.1\angle -25°V$ with port 2 open-circuited (ii) $I_2 = 0.1\angle 0°A$, $V_1 = 3.1\angle -80°V$, $V_2 = 4.2\angle 60°V$, with port 1 open-circuited. Find Z-parameters. (CO6)

(b) Find the Z-parameters and Y-parameters of the following network shown in Fig. 7. (CO6)

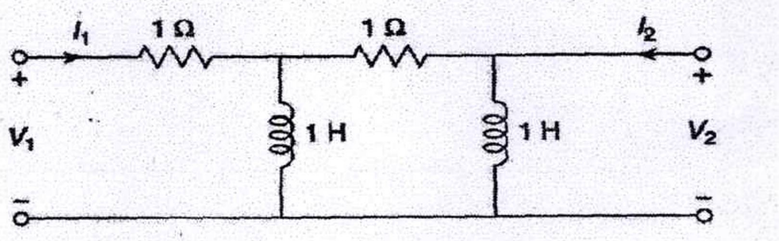

Fig. 7

(c) Determine transmission parameters for the network shown in Fig. 8. (CO6)

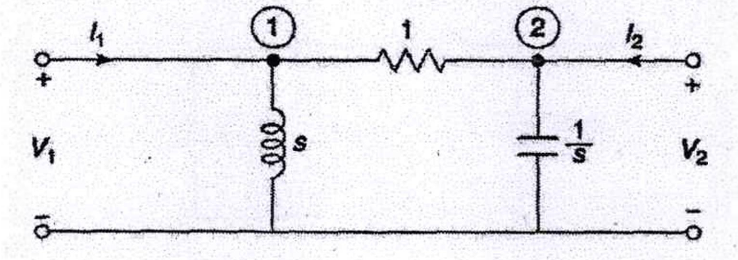

Fig. 8

5. (a) Find the value of R if the average power dissipated in the resistor is 1000 W if the voltage has the following Fourier series : (CO5)

$$v(t) = 200\sin\omega t + 100\sin 3\omega t + 50\sin 5\omega t$$

(b) Find the response voltage in the network shown in Fig. 9. Use Fourier transform method : (CO5)

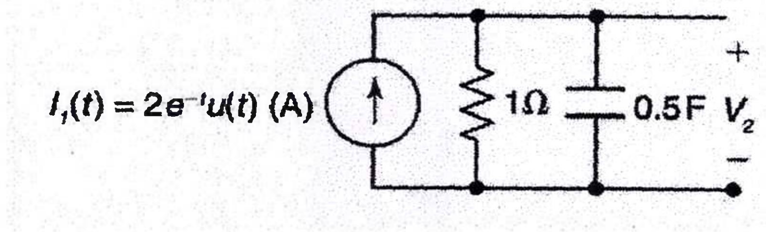

Fig. 9

(c) A complex wave is given by $i(t) = 10 + 100 \sin \omega t + 40 \sin 5\omega t$. Determine the average value and r.m.s. value of the complex wave.

(CO5)

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TEE–302

B. TECH. (EE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

ELECTRICAL MACHINE–I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Across the HV side of single phase 200 V/400 V, 50 Hz transformer, an impedance of $32 + j24\Omega$ is connected, with LV side supplied with rated voltage and frequency. Calculated the supply current and the impedance seen by the supply side. (CO3)

   (b) A transformer having core loss and exciting voltamperes at $B_{max} = 1.5$ T and 60 Hz were found to be $P_{core} = 16$ W, $(VI)_{rms} = 20$ VA and induced voltage was $V_{max} = 242$ when winding had 200 turns, find power factors, core loss current and magnetizing current. (CO3)

   (c) Draw the phasor diagram of transformer at lagging power factor and hence deduce the relation for minimum voltage regulation. (CO3)

2. (a) A 20 kVA 8000/480 V distribution transformer has the following resistance and reactance : (CO2)

$$X_p = 32\ \Omega$$

$$X_s = 45\ \Omega$$

$$R_c = 250\ \text{k}\Omega$$

$$R_p = 0.05\ \Omega$$

$$R_s = 0.06\ \Omega$$

$$X_m = 30\ \text{k}\Omega$$

Find the equivalent circuit of this transformer referred to the high-voltage side. Current in primary as well as secondary.

(b) Discuss open circuit test and short circuit test of transformer with suitable diagram and obtain the condition for maximum efficiency. (CO2)

(c) The equivalent circuit impedances of a 20 kVA, 8000 V/240 V, 60 Hz transformer are to be determined. The open-circuit test and the short-circuit test was performed on the primary side of the transformer, and the following data were taken : (CO2)

Open-circuit test (on primary) $V_{OC} = 8000$ V, $I_{OC} = 0.214$, $V_{OC} = 400$ W

Open-circuit test (on primary) $V_{SC} = 489$ V, $I_{SC} = 2.5$ A, $P_{SC} = 240$ W

Find the impedances of the approximate equivalent circuit referred to the primary side, and sketch that circuit.

3. (a) List and describe the types of losses that occur in a transformer and hence obtain the condition for maximum efficiency. (CO1)

(b) A 100-VA 120/12-V transformer is to be connected so as to form a step-up autotransformer (see Figure 1). A primary voltage of 120 V is applied to the transformer. (CO1)

(i) What is the secondary voltage of t he transformer ?

(b) What is its maximum volt ampere rating in this mode of operation ?

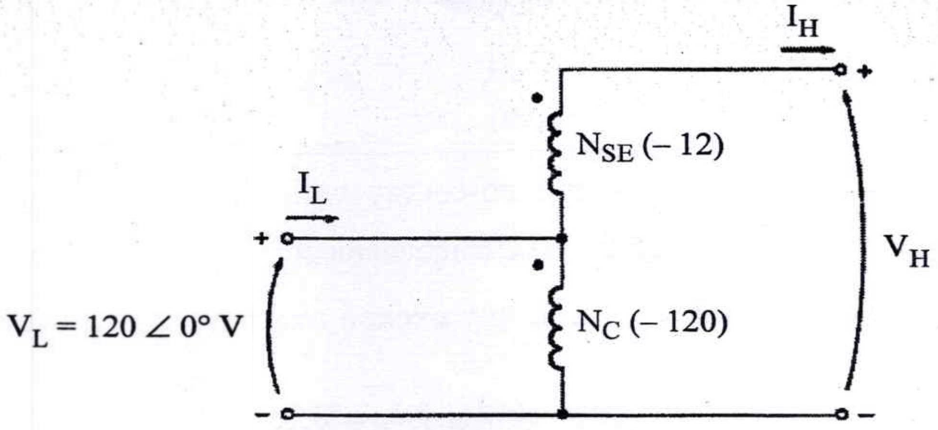

Fig. 1

(c) A wrought iron bar 30 cm long and 2 cm in diameter is bent into a circular shape as shown in fig. 2 below. It is then wound with 600 turns of wire.

Calculate the current required to produce a flux of 0.5 mWb in the magnetic circuit in the following cases : (CO1)

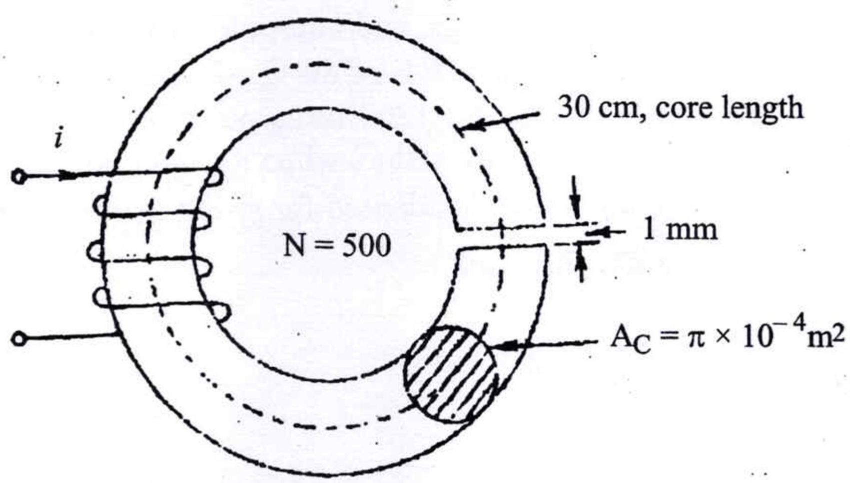

Fig. 2

(i) no air-gap;

(ii) with an air-gap of 1 mm; $\mu_{r\ (iron)} = 4000$ (assumed constant); and

(iii) with an air-gap of 1 mm; assume following data for the magnetization of iron :

H in At/m	**B in T**
2500	1.55
3000	1.59
3500	1.6
4000	1.615

4. (a) Discuss energy and co-energy with suitable diagram, why magnetic field is needed for electromechanical energy conversion. (CO1)

(b) For a system of multiply excited magnetic system various inductances are : (CO1)

$$L_{11} = (4 + \cos 2\theta) \times 10^{-3}\,H, L_{12} = 0.15 \cos\theta\,H, L_{22} = (20 + 5\cos 2\theta)\,H$$

find the torque developed if $i_1 = 1$ A and $i_2 = 0.02$ A

(c) Deduce the relation for co-energy and torque for doubly excited magnetic field system. (CO1)

5. (a) What are various types of d.c. machines based on excitation system ? Explain them with suitable diagram. (CO4)

(b) A 50-hp, 250-V, 1200 r/min d.c. shunt motor with compensating windings has an armature resistance (including the brushes, compensating windings, and interpoles) of 0.06 Ω. Its field circuit has a total resistance R_{adj} + RF of 50 Ω, which produces a no-load speed of 1200 r/min. There are 1200 turns per pole on the shunt field winding. Find the speed of this motors when its input current is 100 A. (CO4)

(c) What are the methods adopted for speed control of d.c. motor ? Discuss with suitable diagram. (CO4)

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TEE–303

B. TECH. (EE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

ELECTRICAL AND ELECTRONIC MEASURING INSTRUMENTS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the terms measurements and measurement system ? Describe the functional element of measurement system with a block diagram. (CO1)

(b) Differentiate between electrical, electronic and mechanical instruments with suitable examples. (CO1)

(c) Explain the following terms : (CO1)

(i) Accuracy

(ii) Selectivity

(iii) Sensitivity

(iv) Errors

P. T. O.

2. (a) Discuss briefly about bridges. Draw the Anderson's bridge circuit and derive necessary equations and explain it. (CO2)

   (b) Maxwell bride is used to measure an inductive impedance the bridge constants at balance are $C_1 = 0.05\ \mu F$, $R_1 = 500\ k\Omega$, $R_2 = 10.1\ k\Omega$ and $R_3 = 10\ k\Omega$. find the series equivalent of the unknown impedance. (CO2)

   (c) How can we increase the range of Ammeter and Voltmeter ? (CO2)

3. (a) Explain different types of instruments transformer. (CO3)

   (b) Explain current transformer and potential transformer with a neat and clean diagram. (CO3)

   (c) Why Kelvin's bridge used ? Explain with diagram. (CO2)

4. (a) Explain digital voltmeter with block diagram. (CO4)

   (b) What is the difference between analog voltmeter and digital voltmeter ? (CO4)

   (c) Write a short note on Flip-flop, Registers and Counters. (CO4)

5. (a) Draw the block diagram of a Cathode ray oscilloscope and explain function of each block in detail of each block in detail. (CO5)

   (b) Explain the major parts of Cathode ray tube with a block diagram. Compare (CO5)

   (c) Explain different types of CRO with block diagram. (CO5)

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TEE–304

B. TECH. (EE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

ELECTROMAGNETIC FIELDS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss cylindrical co-ordinate system. If A_x, A_y, and A_z are the component of vector A in Cartesian co-ordinate system and A_ρ, A_φ, and A_z are component of same vector in cylindrical co-ordinate systems them find the relation between these two components. (CO1)

   (b) Determine the divergence of the vector fields : (CO1)

   (i) $P = \left(x^2yz\right)a_x + (xz)a_z$

   (ii) $Q = (\rho\sin\varphi)a_\rho + \left(\rho^2 z\right)a_\varphi + (z\cos\varphi)a_z$

P. T. O.

(c) Convert a point $P(4,-3,6)$ and a vector $R = z\,a_x + y\,a_z$ into cylindrical coordinate systems. (CO1)

2. (a) Derive the expression of electric field intensity at the perpendicular bisector point of linear charge distribution of charge per unit length 'λ' Assume the length of linear distribution is 'a' and distance of the point is $a/2$ from the center of linear change distribution. (CO2, CO3)

(b) Discuss Gauss' law. Derive Maxwell equation which is based on Gauss' law. (CO2, CO3)

(c) Point charges $1\ \text{mC}$ and $-2\ \text{mC}$ are located at $(3, 2, -1)$ and $(-1, -1, 4)$, respectively. Calculate the electric force on a $10\ \text{nC}$ charge located at $(0, 3, 1)$ and the electric field intensity at that point.

(CO2, CO3)

3. (a) Explain convection and conduction currents. Derive mathematical equations also. Also derive the magnetic vector potential. (CO3, CO4)

(b) Discuss the following : (CO3, CO4)

(i) Ohm's law in point

(ii) Continuity Equation

(c) If potential (V) at any point is given as (i) $V(x, y, z) = 5x^3y^2z$ (ii) $V(\rho, \varphi, z) = z\cos(\varphi)/\rho$, then find change per unit volume at $(-3, 1, 2)$ and $(3, 30°, 2)$. (CO3, CO4)

4. (a) State and explain (i) Biot-Savart's Law (ii) Ampere's circuital law.

(CO5)

(b) A conductor of length 100 cm moves at right angles to uniform field of strength 1T, with velocity of 50 m/s. Calculate the emf induced in it. Also, calculate the value of induced emf when conductor moves at an angle of 30° to the direction of field. (CO5)

(c) Discuss Faraday's law of electromagnetic induction. Derive Maxwell equation using Faraday's law of electromagnetic induction. (CO5)

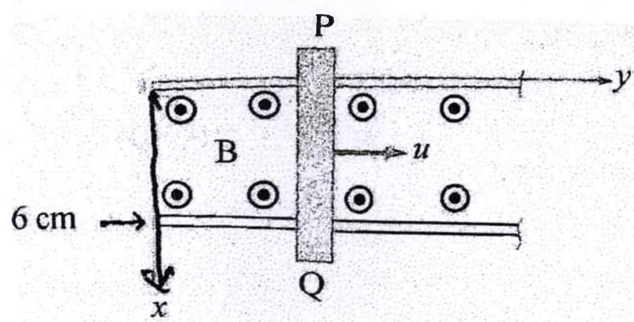
5. (a) Discuss the differential and integral from of Maxwell's equations. (CO6)

(b) Derive the wave equations for conducting medium. (CO6)

(c) A conducting bar can slide freely over two conducting rails as shown in figure. Calculate the induced voltage in the bar. (CO6)

(i) If the bar is stationed at $y = 8$ cm and $B = 4\cos\left(10^6 t\right) a_z \text{mWb/m}^2$

(ii) If the bar slides at a velocity $u = 20a_y$, m/s and $B = 4a_z \text{mWb/m}^2$.



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TEE–305

B. TECH. (EE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

ANALOG ELECTRONICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

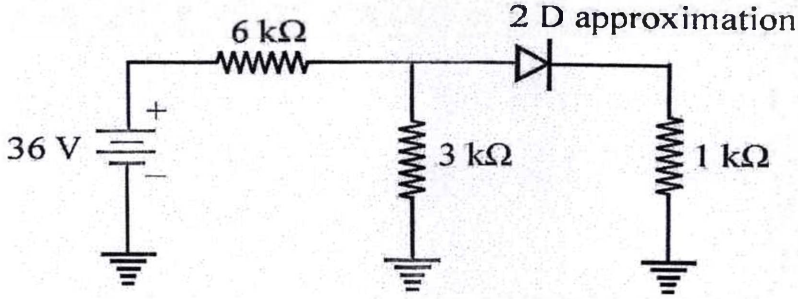
(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Write a brief note on conductors, insulators and semiconductors.

(b) Calculate the load voltage, load current and diode power in the following circuit using second approximation :



P. T. O.

(c) Draw the circuit diagram of a full-wave bridge rectifier. Write its working with the help of wave shapes.

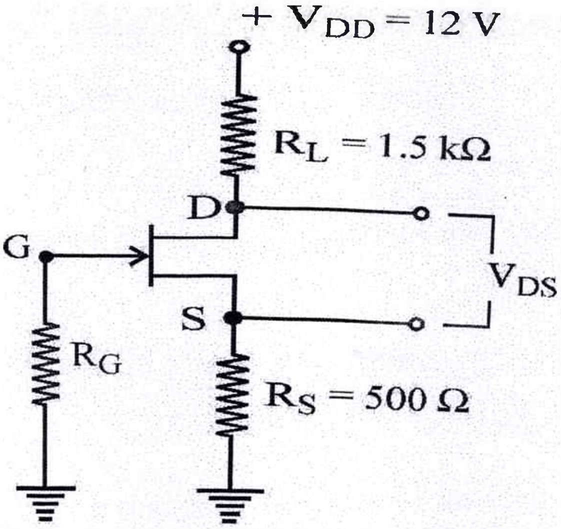
2. (a) Describe the ways of D. C biasing an FET connected in common source configuration.

(b) (i) What are the effects of Gate voltage ?

(ii) Explain what purpose R_G serves.

(c) Find the value of V_{DS} and V_{GS} in the following circuit for $I_D = 4mA$:

(CO)



3. (a) An NPN transistor is connected in common emitter configuration. (CO)

$V_{CC} = 30$ V, $V_{EE} = -30$ V, $R_L = 10$ kΩ, $R_B = 20$ kΩ and $\beta = 100$

Neglect V_{BE}.

Find (i) I_E, (ii) I_B (iii) I_C (iv) V_{CE}

(b) Answer "True" or "False." If the answer is "false," give the right answer :

(i) A bipolar transistor consists of three equally doped regions.

(ii) In a normally-biased NPN transistor, collector is at higher positive potential than base.

(iii) Amongst the three currents of a transistor, I_E is the largest.

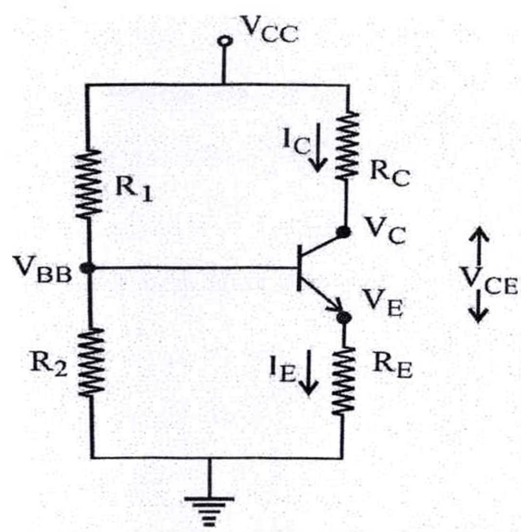
(iv) Theoretically, the CB configuration of a transistor yields the highest current gain.

(v) Temperature- dependent leakage current I_{CO} of a transistor is due to the majority charge carriers.

(vi) When in a circuit $I_C = 0$, the transistor is said to be cut-off.

(vii) A transistor is said to be saturated when $V_{CE} = 0$.

(c) In a voltage- divider bias circuit of an NPN transistor as show below :

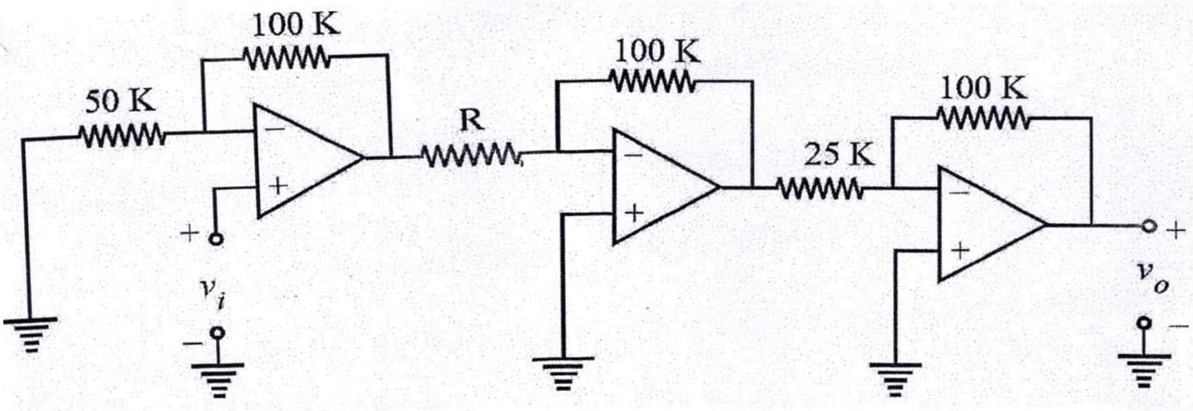


Write the equation for V_{BB}, V_E, I_E, I_C, V_C and V_{CE}.

4. (a) What is a unity Gain Differential Amplifier ? Draw circuit using an Op-Amp, and write its applications.

(b) Draw the circuit of a three input summing Amplifire using an Op-Amp in its inverting configuration. Derive expression for the circuits output V_O.

(c) In the cascade of ideal OP-Amps, as shown below, of $V_i = 2V$ and $V_o = 30\,V$, determine the value of R.



5. (a) What are various types of active filters ? Draw their various circuits employing Operational Amplifiers.

(b) Explain the working of an instrumentation amplifier, employing Operational Amplifiers, with the help of its circuit diagram.

(c) Draw the circuit diagram of a square wave generator and a triangular wave generator. Explain their working.

Roll No. ....

VAC–301

B. TECH. (EE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

BASICS OF MATLAB

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What do you mean by array ? Explain numeric array and character arrays with suitable example. (CO1/CO2)

   (b) Explain the feature of MATLAB. Also discuss its merits and limitations. (CO1/CO2)

   (c) What is simulink ? How do you use it to model and simulate systems ? (CO1/CO2)

2. (a) List some basic plots and graphs of MATLAB. (CO3)

   (b) Discuss the application on the following : (CO3)

   (i) Bus selector

   (ii) Bus creator

   (iii) Scope block used in MATLAB.

(c) Write a MATLAB to draw sinusoidal plots of five different magnitude on the same plane. (CO3)

3. (a) Discuss branching and looping function. (CO6)

(b) Write a MATLAB to generate multiplication table from 1-10. (CO6)

(c) Write a MATLAB code to determine factorial of number from 1 to 10. (CO6)

4. (a) Using switch and case function, write a MATLAB code to determine area and circumference of the circle. Inputs are radius and case through the command window. (CO5)

(b) Generate matrix, 'A', as shown below. For the matrix 'A', what will be the output of the following commands : (CO5)

(i) Sum (diag(A))

(ii) Sum(diag(fliplr(A)))

(iii) A (1, 2) +A (3, 3) +A′(2, 2)

(iv) Sum (A (1 : 3, 2)) +Sum (A (1 : 3, 3)

$$A = \begin{matrix} 20 & 3 & 7 \\ 5 & 5 & 11 \\ 8 & 6 & 7 \end{matrix}$$

(c) Write a program to evaluate the resonant frequency and quality factor of a parallel resonance RLC circuit. The value of R, L, and C are input through the command window. (CO5)

5. (a) Write the MATLAB code to determine the roof of quadratic equation with the following condition : (CO4, CO6)

   (i) If '*a*' is not equal to zero

   (ii) if '*a*' is equal to zero

   The inputs are coefficient of the quadratic equation, i.e., '*a*', '*b*' and 'c'.

   (b) Explain the difference between global and local variables in MATLAB Also. Explain the difference between a script and a function in MATLAB. (CO4, CO6)

   (c) Write a MATLAB code to determine the sum of numbers from 1 to 100. (CO4, CO6)